

24/07/2026

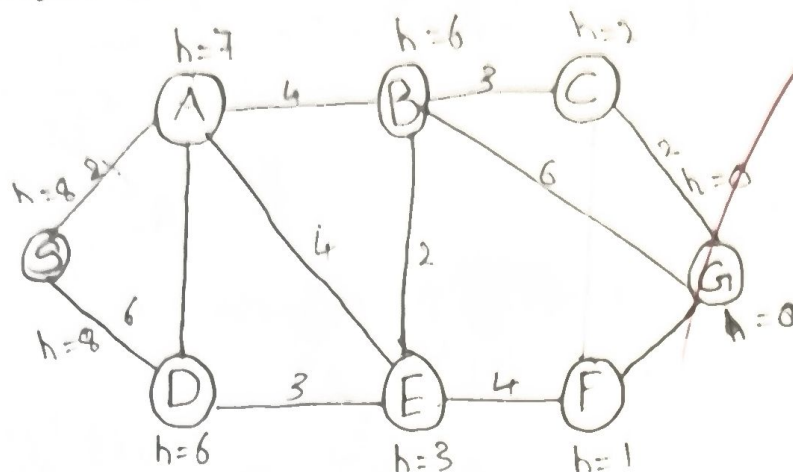
Artificial Intelligence

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CO2 Assessment Tool 2

① Smart Ambulance Navigation :-



② Smart Ambulance Navigation A* search :-

A* Evaluation Function :-A* Uses :-

$$f(n) = g(n) + h(n)$$

Step-1 :- start from S $\rightarrow A = 2$

From A:

$$g(A) = 2$$

$$f(A) = g(A) + h(A) = 2 + 7 = 9$$

So A is selected.

Step-2 :- Expand A

From A:

$$A \rightarrow B = 4$$

$$A \rightarrow D = 6$$

$$A \rightarrow E = 4$$

Step-3 :- Expand E

From E:

$$E \rightarrow B = 2$$

$$E \rightarrow D = 3$$

$$E \rightarrow F = 4$$

$$f(n) = g(n) + h(n)$$

$$f(B) = 8 + 6 = 14$$

$$f(F) = 10 + 1 = 11$$

select F

Step-4 :- Expand F

F connects to G with cost 4.

$$g(G) = 10 + 4 = 14$$

$$f(G) = 14 + 0 = 14$$

But B still has

$$f = 14$$

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Expanding B gives:

$$B \rightarrow C = 3$$

$$g(C) = 2 + 3 = 11$$

$$f(C) = 11 + 2 = 13$$

Then $C \rightarrow G$

$$g(G) = 11 + 2 = 13$$

$$f(G) = 13 + 0 = 13$$

$$B \rightarrow C \rightarrow G$$

$$\text{with cost} = 2 + 4 + 4 + 1 = 11$$

$\therefore$ The minimum ~~same~~ cost is "11"

② Autonomous warehouse Robot :-

5	S	?	?	?	?	?	?
4	?	#	?	?	#	?	?
3	?	?	?	#	?	?	#
2	?	#	?	?	?	#	?
1	?	?	#	?	?	?	G
	1	2	3	4	5	6	7

① Step-1:- Identify the Problem

Start : Charging station.

Goal : Target shelf

Problem : some obstacles are unknown initially

Solution :- use an online search agent

these for:-

$(1, 5) \rightarrow (2, 5) \rightarrow (3, 0) \rightarrow (4, 0) \rightarrow (5, 6) \rightarrow (6, 5) \rightarrow (7, 5)$
 $\rightarrow (7, 4) \rightarrow (6, 4) \rightarrow (6, 3) \rightarrow (5, 3) \rightarrow (5, 2) \rightarrow (5, 1) \rightarrow$
 $(6, 1) \rightarrow (7, 1)$

If $(7, 3)$ is discovered as an obstacle, the robot changes direction and finds another route to $(7, 1)$

online vs offline :-

- * online : unknown obstacles, replans dynamically.
- * offline :- complete map known in advance, plans once.

University Examination Scheduling :- CSP :-

a) CSP Formulation :-

Variables :-

$E_1, E_2, E_3, E_4, E_5, E_6$

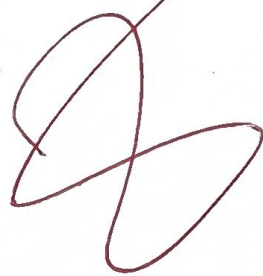
Constraints :-

- $E_1 \neq 2$
- $E_1 \neq E_3$
- $E_2 \neq E_4$
- $E_3 \neq E_5$
- $E_4 \neq E_6$
- $E_2 \neq E_5$

- Start (S) = (1, 5)
- Goal (G) = (7, 1)
- # = obstacle
- ? = unknown cell
- Robot initially knows only some cells.
- It uses A*: $f(n) = g(n) + h(n)$
- when a new obstacle is detected, it updates the map and replans.

A possible new route is:

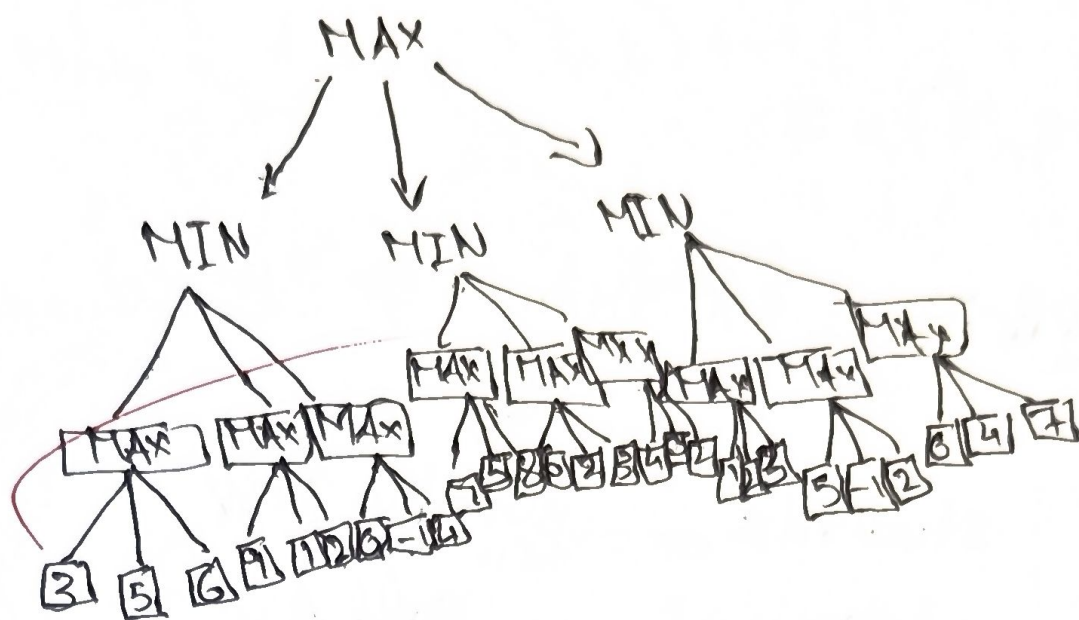
(7, 4)
 ↓
 (6, 4)
 ↓
 (6, 3)
 ↓
 (5, 3)
 ↓
 (5, 2)
 ↓
 (5, 1)
 ↓
 (5, 1)
 ↓
 (7, 1)



However, while moving, the robot may discover another obstacle such as

(6, 2) = blocked.

Game - Playing AI :-



1) Game - Playing AI - Minimax & Alpha-Beta Pruning :-

Minimax - Values :-

$$\bullet \text{ MAX}(3, 5, 6) = 6$$

$$\text{MAX}(9, 2) = 9$$

$$\text{MAX}(0, -1, 4) = 4$$

$$\text{Left MIN} = \text{MIN}(6, 9, 4) = 4$$

$$\bullet \text{ MAX}(7, 5, 8) = 8$$

$$\text{MAX}(6, 2, 3) = 6$$

$$\text{MAX}(4, 0, -2) = 4$$

$$\text{Middle MIN} = \text{MIN}(8, 6, 4) = 4$$

$$\text{MAX}(1, 2, 3) = 3$$

$$\text{MAX}(5, -1, 2) = 5$$

$$\text{MAX}(6, 4, 7) = 7$$

Hall capacity constraints :-

Exam	students	suitable Hall
E ₁	100	Hall A
E ₂	80	Hall A/B
E ₃	70	Hall A/B
E ₄	60	Hall A/B/C
E ₅	90	Hall A/B
E ₆	50	Hall A/B/C

Final Timetable :-

Exam	time slot	students	Hall
E ₁	T ₁	100	Hall A
E ₂	T ₂	80	Hall B
E ₃	T ₂	70	Hall A
E ₄	T ₁	60	Hall C
E ₅	T ₃	90	Hall A
E ₆	T ₄	50	Hall A

Right MIN = $\min(3, 5, 7) = 3$

Therefore,

Root MAX = $\max(2, 4, 3) = 4$

Delivery Route optimization

Method: Hill climbing

1. start with the initial route:

$D \rightarrow A \rightarrow C \rightarrow B \rightarrow E \rightarrow D$

2. calculate its Total distance

3. check the neighbouring routes $s_1 - s_6$

4. choose the route have the Minimum distance

5. From the given routes. The best route is:

$D \rightarrow E \rightarrow C \rightarrow B \rightarrow A \rightarrow D$

6. Its distance is Approximately 23 km.

Final route:-

$D \rightarrow E \rightarrow C \rightarrow B \rightarrow A \rightarrow D$

Total distance: 23 km

	D	A	B	C	E
D	-	6	9	8	7
A	6	-	3	6	6
B	9	3	-	4	5
C	8	5	-	3	5
E	7	6	-	3	-

solution	Route	cost
S ₁	D → A → C → B → E → D	26 km
S ₂	D → B → A → C → E → D	30 km
S ₃	D → A → B → E → C → D	24 km
S ₄	D → E → C → B → A → D	27 km
S ₅	D → A → E → B → C → D	23 km
S ₆	D → C → B → A → E → D	29 km

⇒ first hill climbing Mark

compare 26, 30, 24, 27, 23, 29

23 km

S₇ = D → A → E → B → C → D